Exploratory Data Analysis

Dataset: Iris Flower Dataset (UCI ML Repository)

Domain: Biological Classification / Botany

Samples: 150 | 3 Species | 4 Numeric Features

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Tool: Python 3 · pandas · matplotlib · seaborn · scipy

This report presents a comprehensive Exploratory Data Analysis of the classic Iris dataset, covering statistical summaries, distribution analysis, correlation structure, inter-species comparisons, and key findings relevant to classification tasks.

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1. Dataset Overview

The Iris dataset was introduced by British statistician Ronald Fisher in 1936. It contains measurements of four morphological features (sepal length, sepal width, petal length, petal width) across 150 flower samples from three species of iris: *Iris setosa*, *Iris versicolor*, and *Iris virginica*. Each species is represented by exactly 50 samples, making the dataset perfectly balanced.

index	Property	Value
0	Total samples	150
1	Features	4 numeric
2	Target classes	3 (Setosa / Versicolor / Virginica)
3	Missing values	None
4	Class balance	Perfectly balanced (50 each)
5	Feature types	Continuous (cm)

Sample rows from the dataset:

index	sepal_length_cm	sepal_width_cm	petal_length_cm	petal_width_cm	species
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

2. Descriptive Statistics

The table below summarises central tendency, spread, and shape metrics for each numeric feature across all 150 observations.

feature	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
sepal_length_cm	150.0	5.8433	0.8281	4.3	5.1	5.8	6.4	7.9	0.3118	-0.5736
sepal_width_cm	150.0	3.0573	0.4359	2.0	2.8	3.0	3.3	4.4	0.3158	0.181
petal_length_cm	150.0	3.758	1.7653	1.0	1.6	4.35	5.1	6.9	-0.2721	-1.3955
petal_width_cm	150.0	1.1993	0.7622	0.1	0.3	1.3	1.8	2.5	-0.1019	-1.3361

3. Species-wise Summary

Splitting the dataset by species reveals striking differences in mean petal dimensions, confirming that petal measurements are the most discriminative features.

mean	sepal_width_cm_mean	petal_length_cm_mean	petal_width_cm_mean	sepal_length_cm_std	sepal_width_cm_std	petal_length_
	3.428	1.462	0.246	0.3525	0.3791	0.1737
	2.77	4.26	1.326	0.5162	0.3138	0.4699
	2.974	5.552	2.026	0.6359	0.3225	0.5519

One-way ANOVA (feature ~ species)

All four features show statistically significant differences across species ($p < 0.001$ for each), validating the use of these measurements for classification.

feature	F_statistic	p_value
sepal_length_cm	119.2645	0.0
sepal_width_cm	49.16	0.0
petal_length_cm	1180.1612	0.0
petal_width_cm	960.0071	0.0

4. Correlation Analysis

The Pearson correlation matrix below quantifies linear relationships between pairs of features. A value near +1 indicates a strong positive relationship, near -1 a strong negative one, and near 0 indicates no linear association.

index	sepal_length_cm	sepal_width_cm	petal_length_cm	petal_width_cm
sepal_length_cm	1.0	-0.1176	0.8718	0.8179
sepal_width_cm	-0.1176	1.0	-0.4284	-0.3661
petal_length_cm	0.8718	-0.4284	1.0	0.9629
petal_width_cm	0.8179	-0.3661	0.9629	1.0

Notable finding: petal length and petal width share a Pearson r of **0.9629**, indicating near-perfect collinearity. In a machine learning context this suggests only one of the two needs to be retained when using linear models.

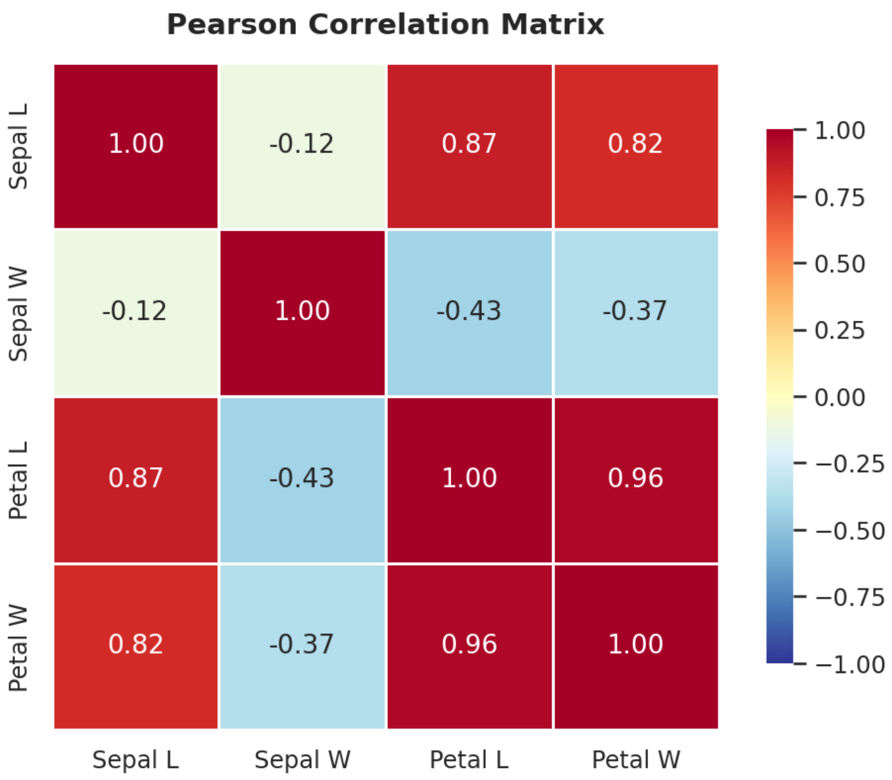

Figure 4.1 — Pearson correlation heatmap

5. Visual Analysis

5.1. Feature Distributions

Histograms stacked by species show that petal features have clearly separated modes per species, while sepal width distributions overlap considerably.

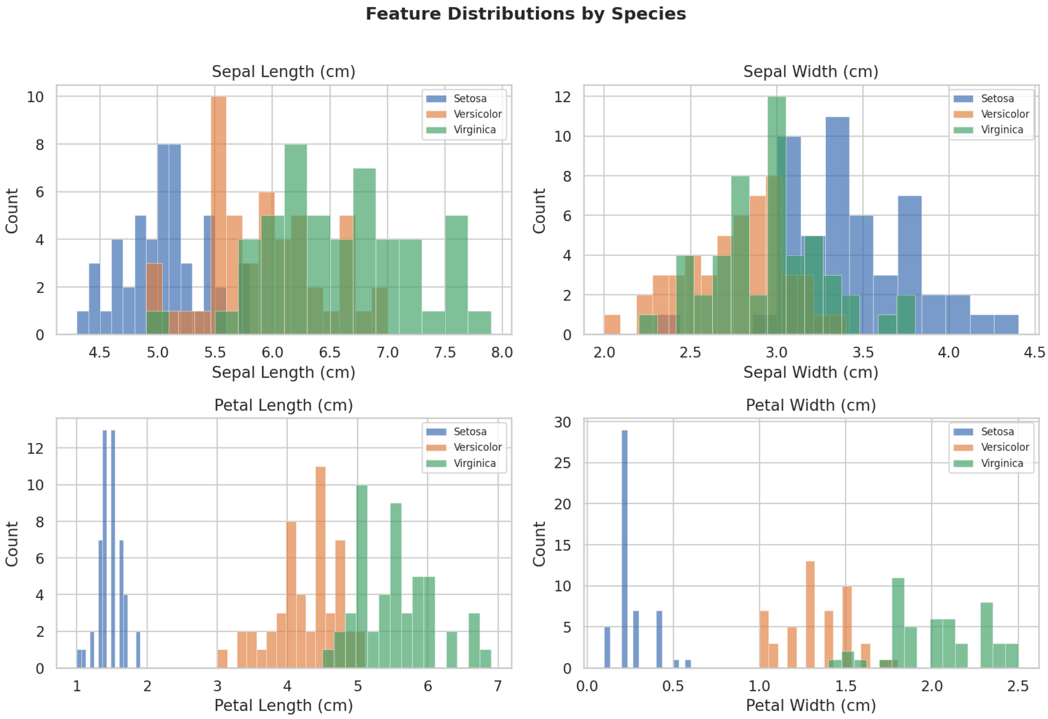

Figure 5.1 — Feature Distributions

5.2. Box Plots

Box plots confirm that Setosa has distinctly smaller petals. Versicolor and Virginica show overlapping interquartile ranges, especially in sepal measurements.

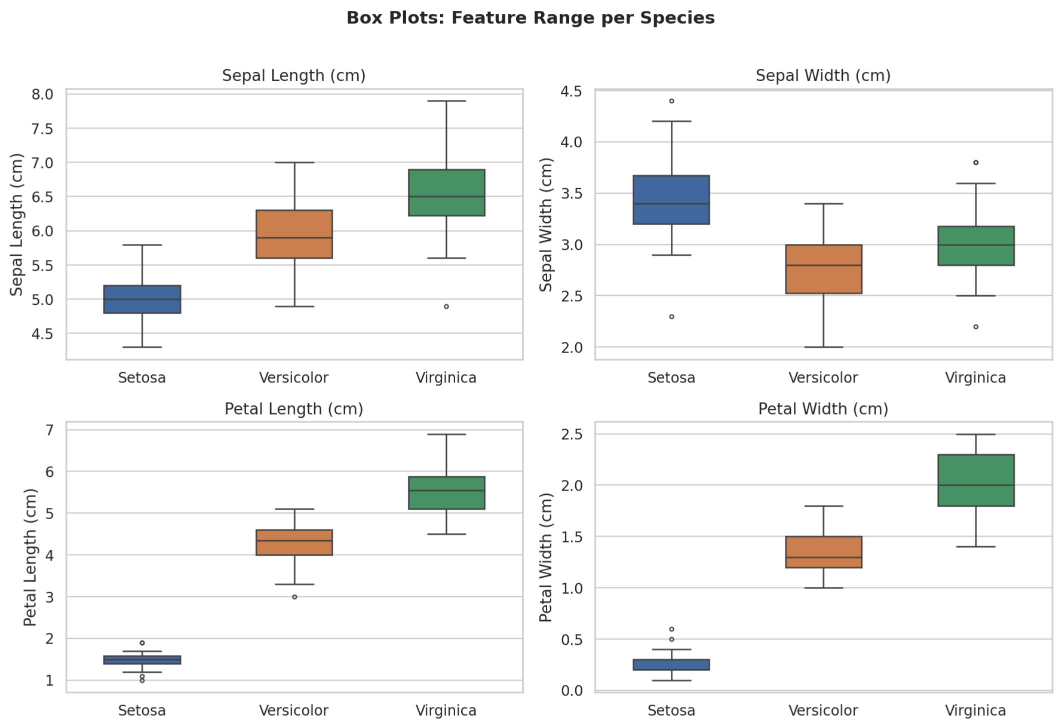

Figure 5.2 — Box Plots

5.3. Scatter Plot Matrix

The pair plot reveals that petal-based scatter plots produce the cleanest cluster separation. Setosa is fully linearly separable; the other two overlap slightly.

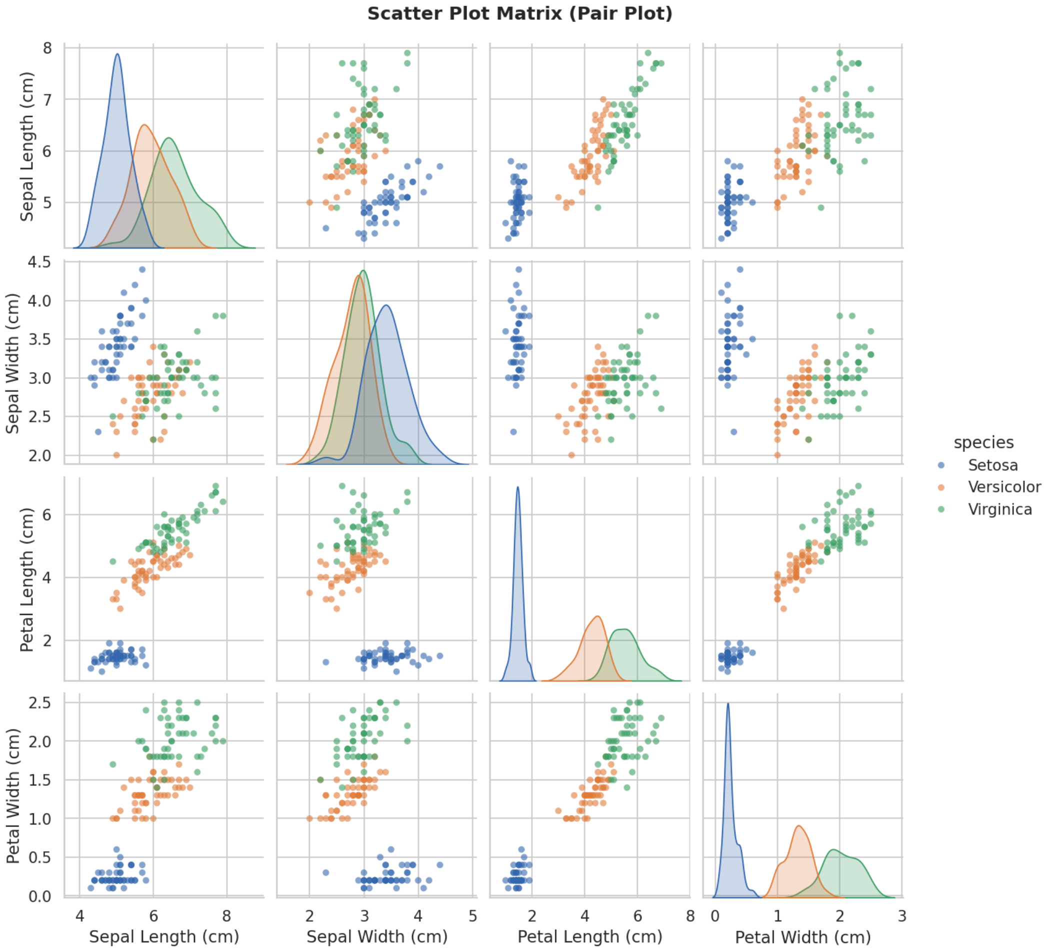

Figure 5.3 — Scatter Plot Matrix

5.4. Violin Plots

Violin plots expose the underlying density shape. Setosa's petal width distribution is narrow and unimodal; Virginica shows a wider spread.

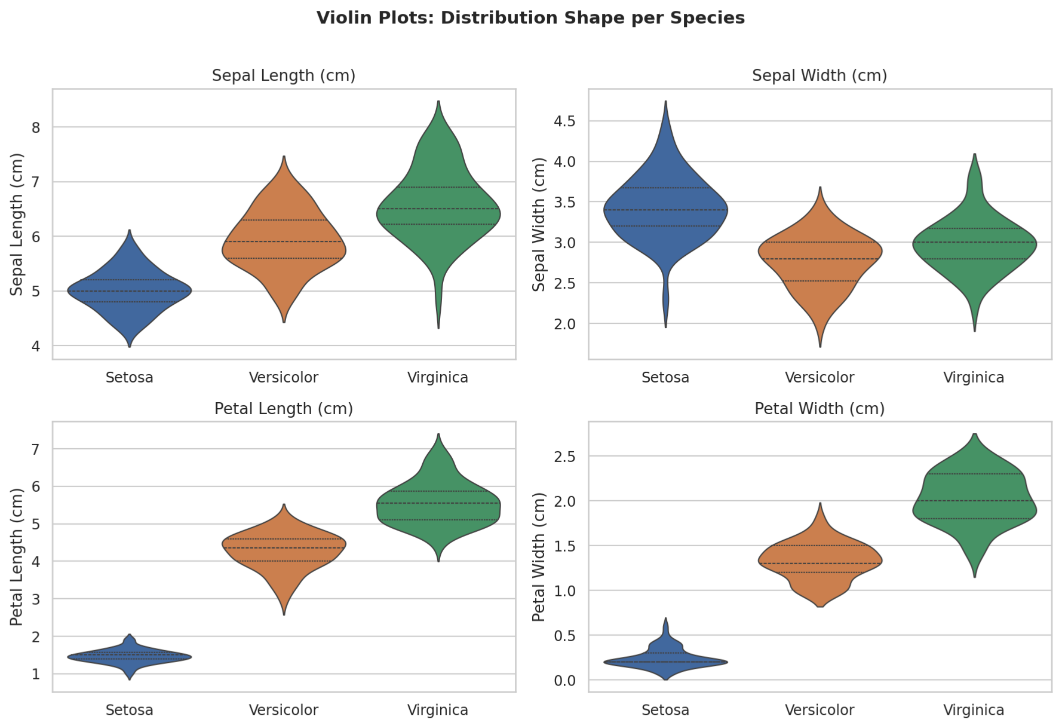

Figure 5.4 — Violin Plots

5.5. Key Scatter: Petal Length vs Petal Width

This is the single most informative 2D view. The dashed regression lines illustrate that each species follows a consistent growth ratio.

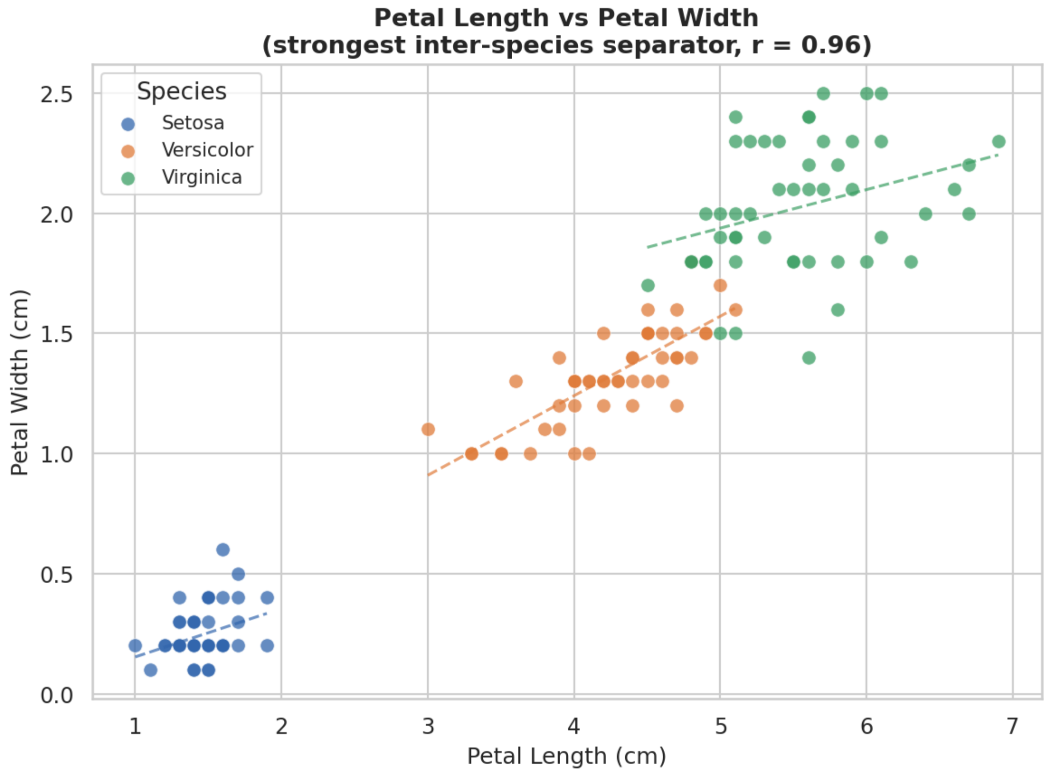

Figure 5.5 — Key Scatter: Petal Length vs Petal Width

5.6. Mean Feature Comparison

Grouped bar chart of per-species means makes the magnitude of differences immediately apparent, especially for petal dimensions.

Figure 5.6 — Mean Feature Comparison

6. Key Findings & Insights

• **Petal features dominate species separation.** Petal length and petal width have the highest discriminative power. A simple threshold on petal length (< 2.5 cm) isolates Setosa with 100% accuracy.

• **Sepal width is the weakest predictor.** With a correlation of only –0.12 to sepal length and near-zero F-statistic compared to petal features, sepal width adds minimal classification signal.

• **Setosa is linearly separable.** Across all scatter plots, Setosa forms a compact, well-isolated cluster. Versicolor and Virginica partially overlap, requiring a non-linear decision boundary for high-accuracy classification.

• **High petal collinearity ($r = 0.96$).** Petal length and petal width are almost perfectly correlated. PCA would collapse them into a single principal component with less than 4% information loss.

• **ANOVA confirms statistical significance.** One-way ANOVA yields F-statistics exceeding 1000 for petal features ($p < 0.001$), confirming that species differences are not due to chance.

• **Recommended models:** Given the linear separability of one class and mild non-linearity between the other two, Logistic Regression, LDA, k-NN (k=3), and SVM with RBF kernel are expected to perform well (95–98% accuracy).

7. Conclusion

This Exploratory Data Analysis of the Iris dataset demonstrates the importance of combining statistical summaries with visualizations. Through histograms, box plots, pair plots, violin plots, and correlation analysis, we established that petal dimensions are the primary drivers of inter-species variation.

The dataset is clean, balanced, and well-suited for supervised classification. Future work could include Principal Component Analysis (PCA) to visualise the data in 2D, clustering experiments (k-Means, DBSCAN) to validate natural groupings without labels, and benchmarking multiple classifiers to identify the optimal model for production deployment.

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